Lecture 7: Other Applications in Finance and Future Trends

Large Language Models in Finance

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Outline

Where We Have Been: A Map of the Book

Ch.	Topic	Primary Technique	Application
1	Text Fundamentals	TF-IDF, word embeddings	Sentiment, classifica
2	Modern LLM Landscape	Transformer, RLHF	General financial NL
3	Training & Fine-Tuning	LoRA, PEFT	Domain adaptation
4	LLM Agents	ReAct, tool use	Research automatio
5	Business Valuation	RAG, summarisation	DCF, comparables
6	Credit Risk	Constrained gen., SHAP	Scoring, fairness
7	**Applications & Future**	**Workflows, multimodal**	**Deployment, gove**

THE BIGGER PICTURE

The through-line. Early chapters: lightweight representations, large corpora, modest compute.

Later chapters: generative and agent-based models, multi-step reasoning.

This chapter: cross-cutting concerns of **deployment, governance, and frontier capability.**

Remaining Application Areas

Portfolio Management

- Natural-language mandate specification $\rightarrow$ quantitative optimisation inputs
- ESG screening from heterogeneous sustainability reports
- Earnings surprise prediction from call transcripts: outperforms bag-of-words dictionaries
- López-Lira (2025): heterogeneous LLM agents in simulated markets $\rightarrow$ real market dynamics + systemic risk from correlated strategies

Regulatory Compliance

- MiFID II suitability report generation (client profile + instrument + regulatory template)
- AML Suspicious Activity Reports: improved consistency, reduced analyst time

Insurance: underwriting risk extraction; claims triage; policy question-answering.

Algorithmic Trading: signal generation (news/filings $\rightarrow$ sentiment

Choosing the Right Deployment Pattern

Pattern	Best for	Data	Latency	C
Zero-shot	General reasoning, novel tasks	None	Low	P
RAG	Document QA, live data, audit trail	Corpus	Medium	C
Fine-tuning	Structured output, consistency	Labelled	Low	H

Three practical heuristics:

1. Prefer RAG over fine-tuning when the task requires grounding in *updatable* documents (current regulation, recent filings). Fine-tuning embeds knowledge statically.
2. Fine-tune when output format is highly structured and consistency across thousands of calls matters more than generality.
3. Zero-shot with a frontier model is a valid baseline before investing in more expensive approaches. FinanceBench: frontier models answer ≈80% of financial QA correctly on the first attempt.

Choosing the Right Deployment Pattern (cont.)

When the selection axes conflict. Real tasks rarely sit cleanly in one column: a task may want both the freshness of RAG and the consistency of fine-tuning, or both low latency and rich generation.

Tiebreaker

When the two selection axes conflict, latency and interpretability dominate: real-time, high-stakes applications default to the simplest, most auditable approach.

Practical consequence. In regulated finance the auditability of a zero-shot or RAG pipeline often outweighs the marginal accuracy gain of a fine-tuned model whose decision logic is harder to document and defend.

Task Property	Enc-only	Dec-only	Enc-Dec	Age
Labelled fine-tuning data available	Strong	Weak	Strong	We
Output is classification / embedding	Strong	Weak	Weak	We
Output requires fluent generation	Weak	Strong	Strong	Stro
Task needs up-to-date external data	Weak	Weak	Weak	Stro
Real-time / low-latency constraint	Strong	Weak	Weak	We
Audit trail / interpretability needed	Strong	Weak	Weak	We
Task spans multiple steps / documents	Weak	Weak	Weak	Stro

Archetypes:

- Trade surveillance classifier $\rightarrow$ encoder-only + fine-tuning
- Regulatory filing summariser $\rightarrow$ decoder-only or enc-dec + RAG
- Autonomous due-diligence agent $\rightarrow$ agentic + zero/few-shot

Financial Workflow as a Directed Acyclic Graph

Definition

A *financial workflow* is a DAG $W = (T, E)$ where:

- $T = \{t_1, \ldots, t_n\}$ is a set of tasks, each $t_i : \mathcal{X}_i \to \mathcal{Y}_i$
- $E \subseteq T \times T$: $(t_i, t_j) \in E$ iff t_j requires an output of t_i
- Acyclic: no directed path from any task back to itself

An *execution* is a topological ordering consistent with E.

Three canonical automation patterns:

Pattern	**Concurrency**	**Best for**
Sequential pipeline	None	Ordered, batch, cascading failures
Event-triggered	Per-event	Semantic triggers, alert systems
Orchestrator-worker	Parallel subtasks	Expert specialisation, replanning

Three Representative Finance Workflows

1. Earnings call processing pipeline (sequential):

1. Retrieve transcript
2. Extract structured guidance (revenue, margin, capex) as JSON
3. Compare against analyst consensus; compute surprises
4. Classify management tone (constructive / cautious / evasive)
5. Draft standardised analyst briefing

Minutes from call end to delivered briefing, vs. hours of analyst time.

2. Regulatory filing monitor (event-triggered): SEC EDGAR RSS feed → classify 8-K by type and materiality → route to appropriate team with summary. Materiality is semantic, not syntactic.

Three Representative Finance Workflows (cont.)

3. Trade surveillance narrative (orchestrator-worker): Rule-based system flags unusual order pattern $\rightarrow$ LLM retrieves news + social media + internal communications $\rightarrow$ drafts structured narrative $\rightarrow$ compliance analyst reviews and decides.

Integration principle

LLMs operate at seconds-to-minutes scale. Execution systems operate at microseconds. Integration is always **asynchronous**. This is structural, not a limitation.

Workflow Patterns: What the Three Examples Share

Common structure across all three:

- A *trigger* (a finished call, an EDGAR feed item, a rule-based flag)
- One or more *LLM stages* doing extraction, classification, or drafting
- A *human checkpoint* before any consequential decision is taken

Why the asynchronous boundary matters. Because LLM stages run at seconds-to-minutes latency, they never sit on the critical path of trade execution. They produce *analysis and narrative*, which humans or downstream systems act on — preserving both speed and auditability.

Financial Research Assistant Architecture

Definition

$\mathcal{A} = (\mathcal{R}, \mathcal{M}, \mathcal{K}, \mathcal{T}, \mathcal{I})$ where:

- $\mathcal{R}$: retrieval engine over corpus $\mathcal{D}$
- $\mathcal{M}$: LLM reader synthesising response from query + retrieved passages
- $\mathcal{K}$: persistent memory (user preferences, portfolio context)
- $\mathcal{T}$: tool layer (market data APIs, filing databases, calculators)
- $\mathcal{I}$: interface layer (text, voice, tables, visualisations)

The five components separate concerns: retrieval, synthesis, memory, tools, and interface can each be developed, validated, and audited independently.

Why Hybrid Retrieval Is Non-Negotiable

Semantic search alone fails on:

- Numerical queries ("what was Apple's FCF in fiscal 2023?")
- Exact-term queries ("Article 22 GDPR", "IFRS 9 ECL")

Dense embeddings capture meaning but blur exact tokens and figures; sparse (BM25) retrieval nails exact terms but misses paraphrase. Production financial assistants run both and fuse the rankings.

UNDER THE HOOD

Reciprocal Rank Fusion:

$$s_{\mathrm{RRF}}(q, d_i) = \frac{1}{k + r_{\mathrm{dense}}(d_i)} + \frac{1}{k + r_{\mathrm{sparse}}(d_i)}, \quad k = 60$$

Parameter-free; robust to different score scales of dense and sparse systems.

RAG Implementation for Financial Documents

Chunking strategy matters. Naive 512-token chunking:

- Fragments financial tables (destroys row/column structure)
- Splits financial statements across chunks
- Disrupts cross-references between filing sections

Section-aware chunking for 10-K filings:

- Each SEC Item indexed independently (Item 1A, 7, 8, etc.)
- Tables extracted as structured objects; indexed by a separate retriever
- Substantial reduction in retrieval noise on financial document QA benchmarks

RAG for Financial Documents: Attribution & Context

Source attribution is a non-negotiable requirement:

- Every claim must cite a retrieved passage
- Claims without retrieved support must be flagged as model inference, not retrieved fact
- In regulated contexts: source attribution = the audit mechanism

Context length paradox (Balogh & Didisheim, 2025)

LLM predictive accuracy follows an inverted-U in context length. Excessive context degrades performance. Mirrors cognitive overload in human analysts.

Implication. More retrieved context is not automatically better: tightly targeted, well-attributed passages beat dumping an entire filing into the prompt.

Sources of Bias in LLM-Based Finance Systems

1. **Training data bias**: frontier LLMs are trained on English-language, US-centric financial content. Geographic bias in risk assessment is a *systematic pricing error*, not merely a fairness concern.
2. **Historical discrimination encoded in data**: LLMs trained on historically discriminatory lending records reproduce those patterns. This is not a model failure — it faithfully reproduces the signal in the training data.
3. **Feedback loop bias**: when LLM outputs influence asset prices, those prices enter future training data, reinforcing the model's priors. Analogous to Soros's reflexivity but at faster timescales. Widespread deployment of similar strategies amplifies volatility.
4. **Anchoring bias**: LLMs weight salient numerical anchors disproportionately. An LLM that just processed a dramatic earnings beat will over-weight that number in subsequent analysis. Documented in human judgment by Tversky and Kahneman (1974); reproduced by LLMs trained on human-generated text.

Fairness Metrics and the Impossibility Theorem

Demographic parity: $\Pr(\hat{Y} = 1 \mid A = a) = \Pr(\hat{Y} = 1 \mid A = b)$ for all a, b.

Equalized odds: $\Pr(\hat{Y} = 1 \mid Y = y, A = a) = \Pr(\hat{Y} = 1 \mid Y = y, A = b)$ for $y \in \{0, 1\}$.

Calibration within group: $\Pr(Y = 1 \mid \hat{P} = p, A = a) = p$ for all p.

UNDER THE HOOD

Chouldechova (2017) Impossibility Theorem. When base rates differ across groups — $\Pr(Y = 1 \mid A = a) \neq \Pr(Y = 1 \mid A = b)$ — a non-trivial classifier **cannot simultaneously** satisfy demographic parity, equalized odds, and calibration within group.

Demographic parity and calibration within group are mutually incompatible whenever base rates differ.

Implication

Fairness is a *family of criteria*, not a single concept. Choosing which

Bias Mitigation: Three Stages

Pre-processing:

- Data rebalancing (oversampling underrepresented groups)
- Causal data augmentation: vary protected attributes while holding other features fixed
- Effect: reduces representation bias; some cost to majority-group accuracy

In-processing:

- Adversarial debiasing: classifier trained to predict outcome while adversary cannot recover sensitive attribute from intermediate representations
- Requires sensitive attribute annotations in training data

Bias Mitigation: Post-Processing and Oversight

Post-processing:

- Threshold adjustment: select decision thresholds satisfying the desired fairness criterion on a held-out calibration set
- Applicable to black-box APIs; comes at the cost of accepting different error rates for different groups

Human-in-the-loop: mandatory for high-stakes decisions regardless of technical mitigation. Regular auditing is essential — bias can emerge or worsen after deployment as the population shifts and models feedback into the features of the next generation.

The Regulatory Landscape for AI in Finance

Framework	Key implications for LLM finance
EU AI Act (2024)	Credit scoring and life insurance risk assessment *high-risk* AI. Conformity assessment before market e try, mandatory registration, transparency obligatio meaningful human oversight.
ECOA / Fair Housing Act	Adverse action notice requirements: specific and acc rate, not generic. In direct tension with LLM opacit
GDPR Art. 22	Right to human review of automated decisions. "V use a large language model" does not constitute mea ingful information about the logic involved.
MiFID II	LLM-generated investment recommendations and m ket analyses must be logged: prompt, model versio output, timestamp.
FSB (2024)	AI amplifies third-party concentration risk, market c relations via herding, and financial disinformation. Ca

SR 11-7 for LLM Systems: Three Lines of Defence

SR 11-7 (2011): when an LLM produces quantitative estimates feeding into material business decisions, it constitutes a model — regardless of its internal architecture.

1. **First line (Development)**: document purpose, scope, data inputs, architecture (foundation model + fine-tuning), known limitations, performance metrics. For LLMs: document failure modes from red-teaming explicitly, even if incomplete.
2. **Second line (Independent Validation)**: adversarial testing including prompt injection, jailbreaking, and distribution shift tests. The FAIR framework (Noguer, 2025) addresses LLM-specific failure modes: temporal data inconsistencies, hallucination in numerical extraction, privacy leakage.
3. **Third line (Internal Audit)**: periodically verify that documentation is current, validation is genuinely independent, findings are tracked to resolution, and model usage has not drifted beyond the validated scope.

SR 11-7 for LLMs: What Changes

The three-lines structure is unchanged from classical model risk management. What changes is the *nature of the risk* the three lines must catch.

What changes for LLMs vs. classical models

The failure mode space is not fully characterised. Emergent capabilities can cause unexpected behaviour on out-of-distribution inputs. Best practice: red-team aggressively and document observed failures honestly.

Consequence for validation. Because failure modes cannot be fully enumerated in advance, validation shifts from one-off sign-off toward continuous, adversarial monitoring that treats every newly observed failure as documentation to be added, not an embarrassment to hide.

Explainability, Monitoring, and Incident Response

Two kinds of explainability:

- **Post-hoc attribution** (SHAP, attention saliency): "what features most influenced this output?"
- **Procedural transparency** (chain-of-thought): provides an auditable reasoning trace, but is a *natural-language reconstruction* of reasoning, not an explanation of neural network internals

Four monitoring signal classes:

1. Input distribution: KL divergence between prompt embeddings detects distributional shift
2. Output quality: accuracy on sampled outputs where ground truth is verifiable
3. Operational: latency, cost, error rates, refusal rates
4. Fairness: periodic re-runs of bias evaluation framework

Incident Severity and Response

Incident severity classification (pre-define before incidents occur):

- Low: Hallucination in low-stakes summarisation $\rightarrow$ standard review cycle
- High: Hallucinated figure in a credit decision $\rightarrow$ immediate escalation, potential reversal, regulatory notification, root-cause investigation

Why severity must be pre-defined. Classifying severity *during* a live incident invites motivated reasoning toward the least disruptive response. Fixing the criteria in advance — and tying each tier to a concrete escalation pathway — is what makes the monitoring signals actionable rather than decorative.

Multimodal LLMs and Long-Context Models

THE BIGGER PICTURE

The limitation of text-only models in finance:

- Earnings presentations contain charts visualising trends
- Annual reports embed tables organising financial statements
- Earnings calls carry prosodic cues (hesitation, urgency) lost in transcription

Multimodal models jointly encode text, images, audio, and structured data. Early benchmark evidence: multimodal models outperform text-only on financial QA tasks requiring chart and table interpretation.

Long-Context Models and the Optimal Architecture

Long-context models (hundreds of thousands of tokens): an entire 10-K can be processed in a single forward pass, linking risk factors to MD&A to legal proceedings without losing context between chunks.

The "lost in the middle" problem

Recall accuracy for long-context models degrades for passages positioned far from either end of the context. Retrieval-augmented approaches do not share this limitation.

Optimal architecture: combine long-context reasoning for intra-document tasks with retrieval for multi-document synthesis.

Autonomous Financial Agents: Opportunities and Risks

The trajectory:

- Quantitative research: hypothesis $\rightarrow$ tested strategy, order-of-magnitude faster
- Trading strategy self-improvement: propose modifications to signal construction, back-test, adopt improvements that pass quality thresholds

Systemic risk concerns (FSB 2024, López-Lira 2025):

- Many market participants deploying similar LLM strategies $\rightarrow$ correlated actions
- Cascade risk: simultaneous repositioning in response to a common signal
- Regulators examining circuit-breaker requirements for AI-driven strategies

Autonomous Agents: Governance Principle

The opportunities and the systemic risks share a root cause: *autonomy*. The governance question is therefore how much autonomy to grant, and over what action space.

Governance principle: three output categories

1. **Informational**: analysis and recommendations
2. **Advisory**: proposed actions for human approval
3. **Executive**: acts on the financial system directly

Confine executive autonomy to pre-approved action spaces with hard limits. Any action outside that space requires explicit human authorisation.

Five Open Research Problems

1. **Calibrated uncertainty quantification**: well-calibrated uncertainty from token-level log-probabilities (Chen et al., 2024) outperforms post-hoc sampling on financial QA. Essential for risk-adjusted decisions and regulatory documentation.
2. **Comparable selection in thin markets**: embedding spaces trained on US financial text are biased toward US market characteristics. Cross-market comparables require market-invariant, firm-sensitive embeddings.
3. **Cross-lingual finance LLMs**: current multilingual models perform substantially worse on non-English regulatory filings (J-GAAP, HGB, local GAAP variants). Corpora and evaluation frameworks do not yet exist.
4. **Feedback loops and memorisation**: Didisheim & López-Lira (2025) formalise the LLM's "predictable memory" — look-ahead bias most severe at low frequencies for large models. Credible identification strategies for loop dynamics are an open problem.
5. **Faithful chain-of-thought verification**: stated reasoning in CoT outputs may not be the actual computation ("unfaithful reasoning"). Verifying consistency between stated reasoning and output has direct regulatory relevance.

The Path Forward: Three Disciplines

THE BIGGER PICTURE

Why capability and responsibility are jointly necessary. An LLM credit model that discriminates unlawfully misallocates capital.
A trading system amplifying volatility damages market infrastructure.
A valuation agent that hallucinates financial figures undermines the professional standards that give financial analysis its social function.

The Path Forward: Three Disciplines

Three disciplines that separate successful deployments from failures:

1. **Measure everything**: retrieval quality, output faithfulness, fairness metrics, distributional drift. What is not measured cannot be managed, and in regulated finance what is not managed creates liability.
2. **Document honestly**: purpose, limitations, known failure conditions, the human oversight structure that catches failures. Documentation is the mechanism by which institutional knowledge survives model updates and team turnover.
3. **Escalate appropriately**: build human review into every consequential decision workflow as a genuine check — not a legal formality — that the reviewer is empowered to override.

The tools are new. The professional standards are not.

Appendix

Extra Material: Code Sketch, Local AI, Course Map

Extra / optional material — beyond the core lecture.

Event-Triggered EDGAR Monitor: Code Sketch

```
CLASSIFICATION_PROMPT = """Classify this newly filed SEC 8-K
    .
Company: {company}
Filing description: {description}

Respond with valid JSON:
{"materiality_score": 1-5, "event_type": "earnings|
    merger_acquisition|...",
 "summary": "..."}"""

def classify_filing(client, company, description):
    message = client.messages.create(
        model="claude-opus-4-5",
        max_tokens=256,
        messages=[{"role": "user", "content":
                   CLASSIFICATION_PROMPT.format(
                       company=company, description=
                           description)}]
    )
    return json.loads(message.content[0].text)
```

Event-Triggered EDGAR Monitor: Code Sketch (cont.)

```
def monitor_edgar(poll_interval=300):
    seen_ids = set()
    while True:
        feed = feedparser.parse(EDGAR_RSS)
        for entry in feed.entries:
            if entry.id not in seen_ids:
                seen_ids.add(entry.id)
                result = classify_filing(client,
                    entry.get("edgar_companyname"),
                    entry.get("summary"))
                if result["materiality_score"] >= 3:
                    dispatch_alert(result)
        time.sleep(poll_interval)
```

OpenClaw: Local Personal AI for Finance Professionals

Design principles:

1. **Privacy by default**: all processing local; financial data never leaves the machine
2. **Meet users where they work**: queries via WhatsApp, Telegram, Slack, Signal
3. **Model flexibility**: cloud (Claude, GPT-4) for capability; local (Llama, Mistral) for sensitivity
4. **Extensibility through skills**: a Python function with a descriptive docstring becomes a callable tool

OpenClaw: Local Personal AI (cont.)

Finance-specific skills:

Skill	What it does
`/earnings AAPL`	Guidance summary vs. consensus from most recent transcript
Filing monitor	Daily digest of new SEC filings from watchlist
Portfolio Q&A	Natural-language questions about local CSV portfolio
Macro digest	Morning briefing from central bank speeches + market moves

Self-modification: the assistant can itself write new skills. "Write me a skill that monitors short interest changes above 5%" $\rightarrow$ new Python file, user reviews, permanently extends capabilities.

Course Summary: The LLM Finance Stack

Layer	**What we have learned**
Text & sentiment	Embeddings, sentiment dictionaries, event extraction
Foundation models	Transformer, attention, pre-training, RLHF, fine-tuning
Domain adaptation	LoRA, PEFT, instruction tuning, training data curation
Agents	ReAct, tool use, orchestrator-worker, multi-agent systems
Business valuation	DCF pipeline, scenario analysis, comparables, Monte Carlo
Credit risk	Constrained generation, calibration, SHAP, fairness, SR 11-7
Workflows	DAG patterns, event triggers, EDGAR integration
Research assistant	Hybrid RAG, persistent memory, OpenClaw
Governance	SR 11-7, EU AI Act, GDPR, MiFID II, incident response
Frontier	Multimodal, long-context, autonomous agents, open problem

The field is young

Researchers and practitioners reading this chapter will shape its norms, its standards, and its failures as much as its successes. Use that privilege deliberately.